

EX. NO: 2

IMPLEMENTATION OF PLAYFAIR CIPHER

AIM:

To write a C program to implement the **Playfair Cipher** encryption technique using a keyword.

ALGORITHM

Step 1: Start the program.

Step 2: Read the keyword from the user.

Step 3: Read the plaintext from the user.

Step 4: Generate a 5×5 key matrix by:

- Removing duplicate letters from the keyword.
- Treating **I** and **J** as the same letter.
- Filling the remaining cells with the unused letters of the alphabet.

Step 5: Prepare the plaintext by:

- Converting all letters to uppercase.
- Replacing **J** with **I**.
- Dividing the plaintext into pairs.
- Inserting **X** between repeated letters in a pair.
- Appending **X** if the plaintext length is odd.

Step 6: Encrypt each pair using the Playfair cipher rules:

- If both letters are in the same row, replace each with the letter to its immediate right.
- If both letters are in the same column, replace each with the letter immediately below it.
- Otherwise, replace each letter with the letter in the same row but in the column of the other letter.

Step 7: Display the generated key matrix and the encrypted ciphertext.

Step 8: Stop the program.

PROGRAM:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

char keyTable[5][5];

int exists(char arr[], int len, char ch)

{

    for(int i = 0; i < len; i++)
```

```

        if(arr[i] == ch)
            return 1;
    return 0;
}

void generateKeyTable(char key[])
{
    char temp[25];
    int k = 0;
    for(int i = 0; key[i] != '\0'; i++)
    {
        char ch = toupper(key[i]);
        if(ch == 'J')
            ch = 'I';
        if(ch >= 'A' && ch <= 'Z' && !exists(temp, k, ch))
            temp[k++] = ch;
    }
    for(char ch = 'A'; ch <= 'Z'; ch++)
    {
        if(ch == 'J')
            continue;
        if(!exists(temp, k, ch))
            temp[k++] = ch;
    }
    k = 0;
    for(int i = 0; i < 5; i++)
        for(int j = 0; j < 5; j++)
            keyTable[i][j] = temp[k++];
}

void findPosition(char ch, int *row, int *col)
{
    if(ch == 'J')
        ch = 'I';

```

```

for(int i = 0; i < 5; i++)
{
    for(int j = 0; j < 5; j++)
    {
        if(keyTable[i][j] == ch)
        {
            *row = i;
            *col = j;
            return;
        }
    }
}

void prepareText(char text[], char prepared[])
{
    int i = 0, k = 0;
    while(text[i] != '\0')
    {
        char a = toupper(text[i]);
        if(a == 'J')
            a = 'I';
        if(!isalpha(a))
        {
            i++;
            continue;
        }
        char b = '\0';
        int j = i + 1;
        while(text[j] != '\0')
        {
            b = toupper(text[j]);

```

```

        if(b == 'J')
            b = 'I';
        if(isalpha(b))
            break;
        j++;
    }
    prepared[k++] = a;
    if(b == '\0')
    {
        prepared[k++] = 'X';
        break;
    }
    else if(a == b)
    {
        prepared[k++] = 'X';
        i++;
    }
    else
    {
        prepared[k++] = b;
        i = j + 1;
    }
}

if(k % 2 != 0)
    prepared[k++] = 'X';

prepared[k] = '\0';
}

void encrypt(char prepared[])
{
    printf("\nCipher Text: ");

```

```

for(int i = 0; prepared[i] != '\0'; i += 2)
{
    int r1, c1, r2, c2;
    findPosition(prepared[i], &r1, &c1);
    findPosition(prepared[i + 1], &r2, &c2);
    if(r1 == r2)
    {
        printf("%c%c",
            keyTable[r1][(c1 + 1) % 5],
            keyTable[r2][(c2 + 1) % 5]);
    }
    else if(c1 == c2)
    {
        printf("%c%c",
            keyTable[(r1 + 1) % 5][c1],
            keyTable[(r2 + 1) % 5][c2]);
    }
    else
    {
        printf("%c%c",
            keyTable[r1][c2],
            keyTable[r2][c1]);
    }
}
printf("\n");
}

int main()
{
    char key[100], text[100], prepared[200];
    printf("Enter Keyword: ");
    scanf("%s", key);

```

```

printf("Enter Plain Text: ");
scanf("%s", text);

generateKeyTable(key);

printf("\nKey Matrix:\n");
for(int i = 0; i < 5; i++)
{
    for(int j = 0; j < 5; j++)
        printf("%c ", keyTable[i][j]);
    printf("\n");
}
prepareText(text, prepared);
printf("\nPrepared Text: %s\n", prepared);

encrypt(prepared);

return 0;
}

```

OUTPUT:

The screenshot shows the Programiz C Online Compiler interface. The code in `main.c` is as follows:

```

164 scanf("%s", key);
165
166 printf("Enter Plain Text: ");
167 scanf("%s", text);
168
169 generateKeyTable(key);
170
171 printf("\nKey Matrix:\n");
172
173 for(int i = 0; i < 5; i++)
174 {
175     for(int j = 0; j < 5; j++)
176         printf("%c ", keyTable[i][j]);
177     printf("\n");
178 }
179
180 prepareText(text, prepared);
181
182 printf("\nPrepared Text: %s\n", prepared);
183
184 encrypt(prepared);
185
186 return 0;
187 }

```

The output of the program is shown on the right:

```

Enter Keyword: pallavi
Enter Plain Text: good girl

Key Matrix:
P A L V I
B C D E F
G H K M N
O Q R S T
U W X Y Z

Prepared Text: G00D

Cipher Text: OURB

=== Code Execution Successful ===

```

The bottom of the image shows a Windows taskbar with the date 03-08-2026 and time 08:46 PM.